

MAJOR PROJECT REPORT

# Seasonal Agriculture Performance Analysis

Data Analytics Project

| Student      | Ranadeep Das                             |
|--------------|------------------------------------------|
| AICTE STU ID | STU6a5f732ab94b61784640298               |
| Dataset      | Seasonal Agriculture Performance Dataset |
| Records      | 4,000                                    |

A concise analysis of seasonal yield, profitability, environmental conditions, crop performance and irrigation.

# 1. Introduction

Agricultural performance changes with season because environmental conditions, farming practices, resource availability and market conditions also change. This project uses data analysis to make those seasonal differences easier to understand.

# 2. Problem Statement

The raw agricultural dataset contains information about farming conditions, production and economic outcomes, but the important seasonal patterns are not immediately visible. The task is to compare seasons and identify meaningful trends, relationships and variations.

# 3. Objectives

- Explore and understand the dataset.
- Clean missing and duplicate data before analysis.
- Compare yield and profitability across seasons.
- Study crop and irrigation performance.
- Examine environmental and resource-use patterns.
- Present evidence-based findings and recommendations.

# 4. Dataset

The dataset contains 4,000 records and 28 variables, covering crops, states, seasons, farm conditions, inputs, irrigation, yield, production, revenue, cost, profit, water use and disease/pest risk.

| Area        | Examples                                               |
|-------------|--------------------------------------------------------|
| Environment | Rainfall, temperature, humidity, soil moisture         |
| Farm inputs | Nitrogen, phosphorus, potassium, fertilizer, pesticide |
| Performance | Yield, production, revenue, profit                     |
| Resources   | Irrigation method, water used, water efficiency        |

## 5. Data Preparation

Before analysis, the dataset was inspected for missing values, duplicates, data types and category distributions. The original data contained 120 missing cells and 0 duplicate rows.

| Variable          | Missing values | Treatment |
|-------------------|----------------|-----------|
| Rainfall_mm       | 48             | Median    |
| Soil_Moisture_pct | 40             | Median    |
| Yield_Tonnes_Ha   | 32             | Median    |

Median imputation was used for the selected numeric fields. This allowed the analysis to retain all 4,000 records.

## 6. Analytical Approach

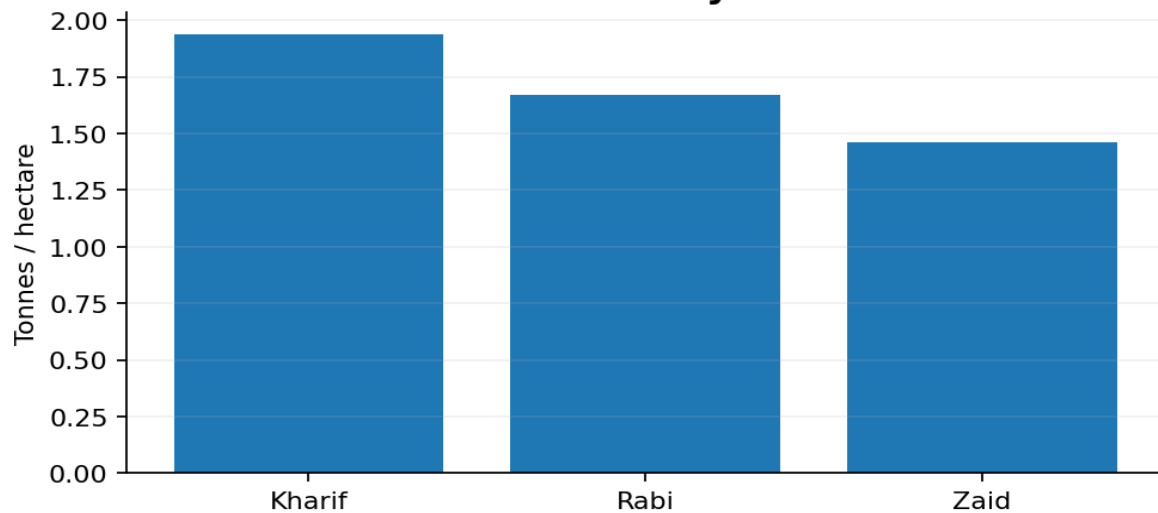
The workflow was kept simple and reproducible: inspect → clean → summarize → compare → visualize → interpret. Median values were preferred for several performance comparisons because they are less affected by extreme observations.

Tools: Python, Pandas, NumPy, Matplotlib and Jupyter Notebook.

## 7. Seasonal Performance

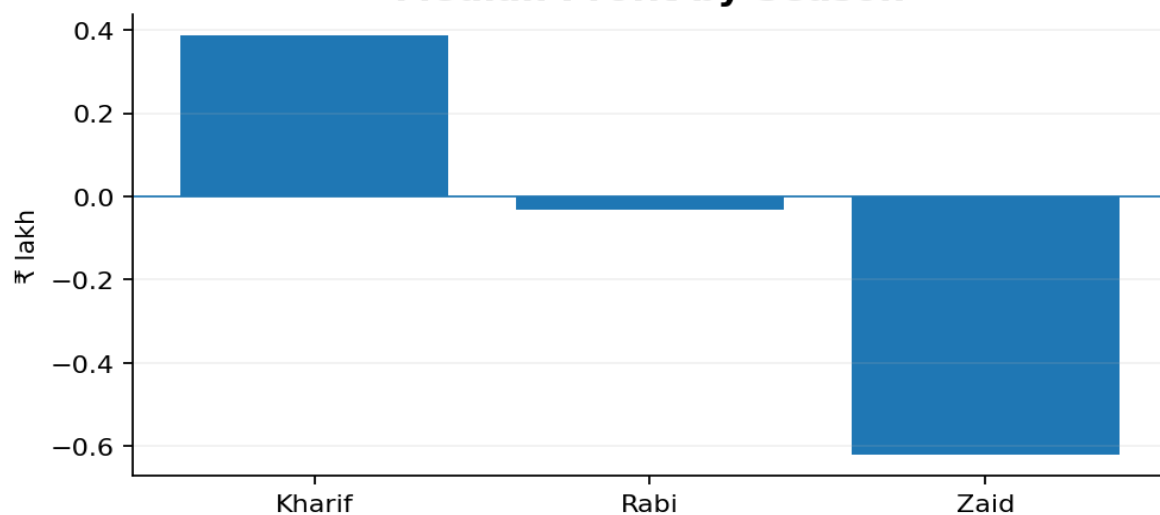
| Season | Farms | Median yield | Median profit | Profitable farms |
|--------|-------|--------------|---------------|------------------|
| Kharif | 1,779 | 1.94 t/ha    | ₹0.39L        | 57.8%            |
| Rabi   | 1,627 | 1.67 t/ha    | ₹-0.03L       | 48.9%            |
| Zaid   | 594   | 1.46 t/ha    | ₹-0.62L       | 35.5%            |

**Median Yield by Season**

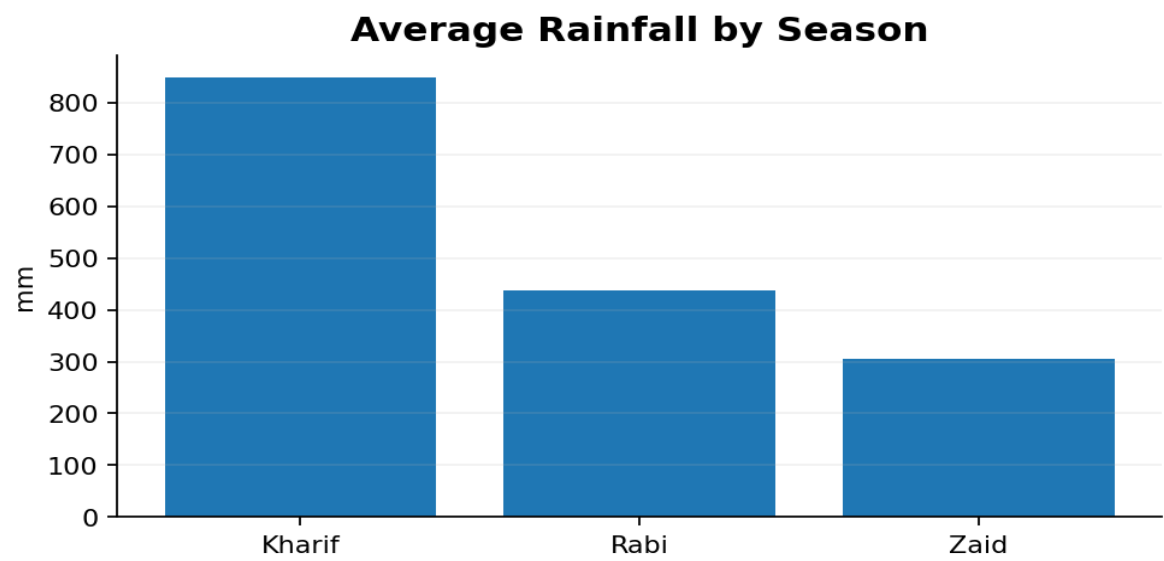


Observation: Kharif has the highest median yield (1.94 t/ha) and the strongest profitability. Zaid records the lowest median profit (₹-0.62 lakh).

**Median Profit by Season**



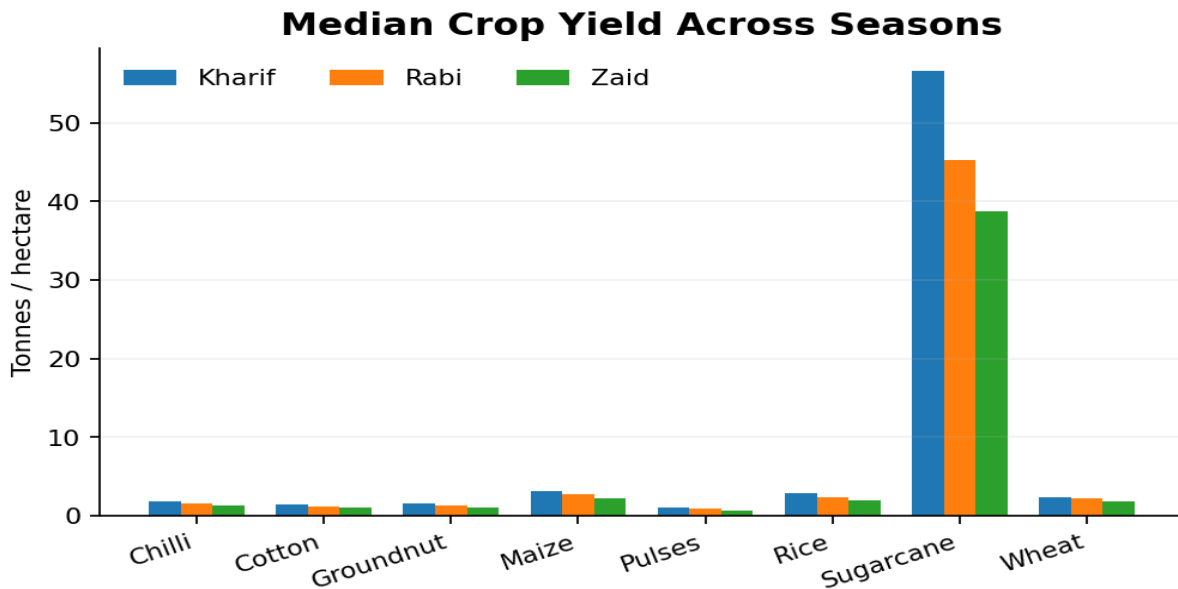
## 8. Environmental Conditions



| Season | Rainfall | Soil moisture | Temperature | Humidity | Pest risk |
|--------|----------|---------------|-------------|----------|-----------|
| Kharif | 849 mm   | 31.1%         | 28.5 °C     | 71.8%    | 54.5%     |
| Rabi   | 438 mm   | 24.1%         | 23.5 °C     | 57.9%    | 40.5%     |
| Zaid   | 305 mm   | 19.3%         | 31.0 °C     | 52.0%    | 38.2%     |

Kharif is much wetter and has higher soil moisture than the other seasons. Zaid is comparatively drier. These differences provide useful context for the seasonal performance results.

## 9. Crop-Level Analysis



The crop comparison shows that performance is not uniform across crops or seasons. Sugarcane is a standout high-yield and high-profit category in the supplied data, while Chilli also shows comparatively strong profitability.

### Top crop summary

| Crop      | Farms | Median yield | Median profit |
|-----------|-------|--------------|---------------|
| Sugarcane | 305   | 47.88 t/ha   | ₹6.91L        |
| Maize     | 551   | 2.84 t/ha    | ₹-0.43L       |
| Rice      | 690   | 2.44 t/ha    | ₹-0.58L       |
| Wheat     | 614   | 2.21 t/ha    | ₹-0.76L       |
| Chilli    | 412   | 1.56 t/ha    | ₹5.36L        |
| Groundnut | 424   | 1.40 t/ha    | ₹0.36L        |
| Cotton    | 508   | 1.25 t/ha    | ₹0.62L        |
| Pulses    | 496   | 0.94 t/ha    | ₹0.03L        |

# 10. Irrigation and Water Use

Irrigation methods were compared using median yield, median profit and average water efficiency.

| Method    | Farms | Median yield | Median profit | Water efficiency |
|-----------|-------|--------------|---------------|------------------|
| Drip      | 915   | 1.94 t/ha    | ₹0.53L        | 6.27             |
| Sprinkler | 734   | 1.79 t/ha    | ₹0.08L        | 4.67             |
| Rainfed   | 1,041 | 1.72 t/ha    | ₹-0.10L       | 7.56             |
| Flood     | 1,310 | 1.67 t/ha    | ₹-0.12L       | 3.44             |

Drip irrigation has the highest median yield among the irrigation methods in this dataset. Yield and water efficiency should still be considered together rather than using yield alone.

# 11. State-Level Comparison

| State          | Farms | Median yield | Median profit |
|----------------|-------|--------------|---------------|
| Telangana      | 516   | 1.83 t/ha    | ₹0.16L        |
| Maharashtra    | 512   | 1.77 t/ha    | ₹-0.01L       |
| Karnataka      | 489   | 1.75 t/ha    | ₹0.15L        |
| Punjab         | 484   | 1.74 t/ha    | ₹0.12L        |
| Tamil Nadu     | 485   | 1.74 t/ha    | ₹-0.02L       |
| Gujarat        | 488   | 1.73 t/ha    | ₹0.04L        |
| Madhya Pradesh | 497   | 1.69 t/ha    | ₹-0.04L       |
| Andhra Pradesh | 529   | 1.67 t/ha    | ₹-0.04L       |

## 12. Relationship Analysis

The correlation between yield and profit is 0.488, showing a positive linear association in this dataset. Correlation does not prove causation.

| Variable                      | Correlation with Profit |
|-------------------------------|-------------------------|
| Revenue_INR                   | 0.887                   |
| Production_Tonnes             | 0.554                   |
| Water_Efficiency_t_per_1000m3 | 0.489                   |
| Yield_Tonnes_Ha               | 0.488                   |
| Market_Price_INR_Tonne        | 0.225                   |
| Water_Used_m3                 | 0.190                   |
| Farm_Area_Hectares            | 0.115                   |

This analysis is descriptive. More advanced modelling would be needed to explain or predict outcomes.



## 13. Key Findings

- Kharif has the highest median yield at 1.94 t/ha.
- Kharif has the highest median profit at ₹0.39 lakh.
- Zaid has the lowest median profit at ₹-0.62 lakh.
- Seasonal environmental conditions differ clearly, especially rainfall and soil moisture.
- Drip irrigation records the highest median yield among the listed methods.
- Sugarcane is a standout high-yield/high-profit crop category.
- Yield and profit have a positive association in the dataset.

## 14. Recommendations

- Use seasonal performance when planning crop and resource allocation.
- Pay closer attention to cost and profitability during weaker seasons.
- Evaluate irrigation using both yield and water efficiency.
- Use crop-specific seasonal performance for crop planning.
- Combine farm records with weather and market-price information.
- Validate these observations with field-level data before operational use.

## 15. Limitations

- The dataset is observational; causal relationships cannot be established.
- There is no long time-series dimension for detailed year-to-year analysis.
- Selected missing numeric values were median-imputed.
- External weather forecasts and market shocks are not modelled separately.
- Findings are specific to the supplied dataset.

## 16. Future Scope

- Add time-based data for longer-term seasonal trends.
- Build an interactive dashboard with filters for season, crop, state and irrigation.
- Develop regression or machine-learning models for yield and profit prediction.
- Integrate weather forecasts and market prices.
- Validate findings with real farm-level data.

## 17. Conclusion

The analysis shows clear seasonal differences in agricultural performance. Kharif provides the strongest overall combination of yield and profitability in the supplied data, while Zaid shows weaker economic outcomes. Crop and irrigation comparisons add further context.

The project demonstrates how a structured data-analysis workflow can convert raw agricultural records into concise, evidence-based insights.

## 18. Project Information

| Student      | Ranadeep Das                                                    |
|--------------|-----------------------------------------------------------------|
| AICTE STU ID | STU6a5f732ab94b61784640298                                      |
| Notebook     | Seasonal_Agriculture_Performance_Analysis.ipynb                 |
| Presentation | Ranadeep_Das_Seasonal_Agriculture_Performance_Analysis_PPT.pptx |